Lecture-22 Scribe: Inequalities and Tail Bounds (From Markov's Inequality Onward)

(Exam Preparation Notes – Step-by-Step Reasoning)

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1. Markov's Inequality

1.1 Definition

Let X be a non-negative random variable ($X \geq 0$).

For any $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

1.2 Step-by-Step Proof

Step 1: Start from expectation

$$E[X] = \sum_x x \cdot P(X = x)$$

Step 2: Split based on threshold a

$$E[X] = \sum_{x<a} xP(X = x) + \sum_{x \geq a} xP(X = x)$$

Step 3: Bound the large-value part
For all $x \geq a$,

$$x \geq a \Rightarrow xP(X = x) \geq aP(X = x)$$

Thus,

$$\sum_{x \geq a} xP(X = x) \geq a \sum_{x \geq a} P(X = x)$$

$$= a \cdot P(X \geq a)$$

Step 4: Combine

$$E[X] \geq a \cdot P(X \geq a)$$

Step 5: Rearranging

$$P(X \geq a) \leq \frac{E[X]}{a}$$

1.3 Example

If:

$$E[X] = 10$$

Then:

$$P(X \geq 50) \leq \frac{10}{50} = 0.2$$

1.4 Key Insight

No assumption on distribution.
Only requires non-negativity.
Provides a loose but universal bound.

2. Chebyshev's Inequality

2.1 Motivation

Markov's inequality only uses expectation.
To get a tighter bound, we incorporate variance.

2.2 Definition

For any random variable X with mean μ and variance σ^2,

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

2.3 Proof Using Markov's Inequality

Step 1: Define new variable
Let:

$$Y = (X - \mu)^2$$

$Y \geq 0$, so Markov applies.
Step 2: Apply Markov

$$P(Y \geq a^2) \leq \frac{E[Y]}{a^2}$$

Step 3: Substitute back

$$P((X-\mu)^2 \geq a^2) = P(|X-\mu| \geq a)$$

Step 4: Use variance definition

$$E[Y] = E[(X-\mu)^2] = \sigma^2$$

Final Result

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

2.4 Example

If:

$$\mu = 100, \quad \sigma^2 = 25$$

Then:

$$P(|X-100| \geq 10) \leq \frac{25}{100} = 0.25$$

2.5 Insight

Uses both mean and spread.
Stronger than Markov.
Still distribution-independent.

3. Chernoff Bound (Poisson Tail)

3.1 Motivation

Markov and Chebyshev give loose bounds.
Need exponentially decreasing bounds for sums of random variables.

3.2 Idea

Instead of applying Markov to X, apply it to:

$$e^{tX}$$

This amplifies large deviations.

3.3 Markov on Exponential

Apply Markov:

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

3.4 Resulting Bound

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Choose t to minimize the bound.

3.5 Interpretation

Uses moment generating function (MGF).
Produces exponentially tight bounds.
Especially useful for:
Sum of independent random variables
Poisson-type tails

3.6 Key Insight

Transforming the variable makes rare events more visible.
Leads to much stronger guarantees than Chebyshev.

4. Jensen's Inequality

4.1 Convex Functions

A function f is convex if:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

for $0 \leq \lambda \leq 1$.

4.2 Definition of Jensen's Inequality

For a convex function f:

$$f(E[X]) \leq E[f(X)]$$

4.3 Proof Idea (Step-by-Step Reasoning)

Step 1: Use convexity
Convexity says function lies below line segments.
Step 2: Represent expectation as weighted sum

$$E[X] = \sum_i p_i x_i$$

Step 3: Apply convexity repeatedly

$$f\left(\sum_i p_i x_i\right) \leq \sum_i p_i f(x_i)$$

Step 4: Recognize expectation

$$f(E[X]) \leq E[f(X)]$$

4.4 Example Insight

If $f(x) = x^2$ (convex):

$$(E[X])^2 \leq E[X^2]$$

4.5 Interpretation

Applying function after expectation gives smaller value than:
Taking expectation after applying function

5. Case Study: Worst-Case Server Demand

5.1 Problem Context

Predict maximum load on a server.
Exact distribution unknown.

5.2 Approach Using Inequalities

Step 1: Use expectation
Apply Markov for rough bound.
Step 2: Use variance
Apply Chebyshev for tighter bound.
Step 3: Use exponential bounds
Apply Chernoff for strong guarantees.

5.3 Key Insight

These inequalities allow:

Bounding rare events

Designing systems with guarantees

Understanding worst-case scenarios

6. Final Summary (Exam Revision)

Inequality	Uses	Strength
Markov	Expectation	Weak
Chebyshev	Mean + Variance	Moderate
Chernoff	Exponential moments	Strong
Jensen	Convexity	Structural

Final Takeaway

All inequalities aim to bound:

$$P(\text{large deviation})$$

Increasing strength:

Markov $\rightarrow$ Chebyshev $\rightarrow$ Chernoff

Jensen provides a fundamental tool for expectation transformations.